James Madison University

Information Technology (IT) Program

IT 215 Intro to Telecom, Networking & Security

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Lab 4 - Advanced Network Protocols

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19:36:58

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Submitted to:
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Honor Pledge: I have neither given nor received help on this lab that violates the spirit of the JMU Honor Code.

Victoria Liu	Dustin Crawford
Signature	*Signature*
Date	*Date*

Contents

1 Introduction 4
2 Lab Exercises : Advanced Network Protocols 4
  2.1 Exercise 1 - Hosts and Network Devices 4
  2.2 Exercise 2 - Wireless LAN Setup, Iperf Measurement, and Wireshark Analysis 11
3 Conclusion 15
  3.1 Summary 15
  3.2 Expectancy-Value-Cost 15
  3.3 Suggestions 15
  3.4 Best Practices 15
4 Lab References 15
5 Acknowledgments 16

List of Figures

1 IP data via ipconfig command in linux terminal 4
2 IP data via ipcalc command in linux terminal 5
3 IP address acquired from the AP 5
4 AP Login through my browser 6
5 Changed Local IP Address 6
6 New IP address details in ipcalc 7
7 New Network Name and Wireless Channel 7
8 New Specific IP Address 8
9 Failed External Ping 8
10 WAN IP Information 9
11 Successful ping to google.com 10
12 Connection Diagram 10
13 IP Addressed for Connected Machines 11
14 IP Configuration Diagram 12
15 TCP and UDP iperf results 13
16 Wireshark Annotations Pt.1 14
17 Wireshark Annotations Pt.2 14

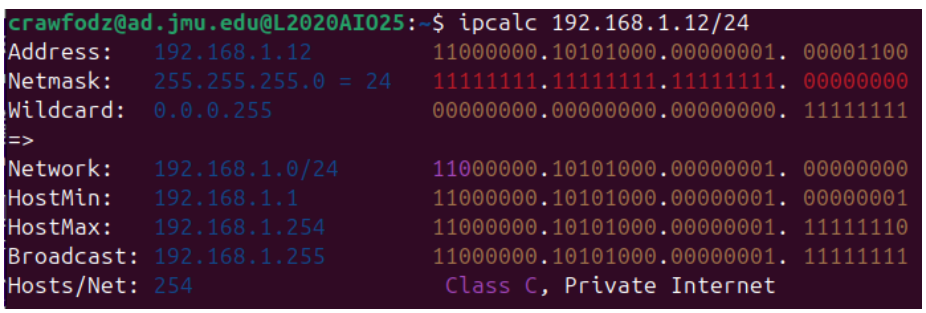
1 Introduction

This lab is about being able to distinguish between different kinds of network devices, understanding the different parts of a network for physical and virtual devices, testing client-server communication, examining packet exchanges in Wireshark, and understanding error detection.

2 Lab Exercises : Advanced Network Protocols

2.1 Exercise 1 - Hosts and Network Devices

T1: Open a Linux terminal on your Ubuntu Host or Linux Mint VM and type ipcalc 192.168.1.12/24. Note that /24 is a shorthand for 255.255.255.0 (it represents a subnet mask of 24 1s). Also, note that 255 in binary is 1111 1111. The response should look like Figure 1 below. As you can see in Figure 1, 192.168.1.12/24 is a private IP address. If the "Private Internet" is not there, then the IP address in public IP address. Here are the results of running that IP in ipcalc.



```
crawfodz@ad.jmu.edu@L2020AIO25:~$ ipcalc 192.168.1.12/24
Address:   192.168.1.12          11000000.10101000.00000001. 00001100
Netmask:   255.255.255.0 = 24    11111111.11111111.11111111. 00000000
Wildcard:  0.0.0.255             00000000.00000000.00000000. 11111111
=>
Network:   192.168.1.0/24        11000000.10101000.00000001. 00000000
HostMin:   192.168.1.1           11000000.10101000.00000001. 00000001
HostMax:   192.168.1.254         11000000.10101000.00000001. 11111110
Broadcast: 192.168.1.255         11000000.10101000.00000001. 11111111
Hosts/Net: 254                    Class C, Private Internet
```

Figure 1: IP data via ipconfig command in linux terminal

Q1: Use ipcalc or an online tool to determine whether 134.126.10.64/22 is a private or public IP address. Explain your reasoning and include a screenshot.

A1: The IP 134.126.10.64/22 is a public address. You can tell because in the Host/Net column in the information after running ipcalc, there is nothing after it says Class B. Since there is nothing, that means it is public. If it were private, it would say "Private Internet" after the class. See: Figure 2.

```
crawfodz@ad.jmu.edu@L2020AIO25:~$ ipcalc 134.126.10.64/22
Address:   134.126.10.64        10000110.01111110.000010 10.01000000
Netmask:   255.255.252.0 = 22   11111111.11111111.111111 00.00000000
Wildcard:  0.0.3.255            00000000.00000000.000000 11.11111111
=>
Network:   134.126.8.0/22       10000110.01111110.000010 00.00000000
HostMin:   134.126.8.1          10000110.01111110.000010 00.00000001
HostMax:   134.126.11.254       10000110.01111110.000010 11.11111110
Broadcast: 134.126.11.255       10000110.01111110.000010 11.11111111
Hosts/Net: 1022                  Class B

crawfodz@ad.jmu.edu@L2020AIO25:~$
```

Figure 2: IP data via ipcalc command in linux terminal

T2: DHCP and NAT Investigation

Q1: State the exact IP address acquired by your Ubuntu machine. identify the network service on the AP that assigned this IP address (DHCP server).

A1: The IP address acquired by the AP was 192.168.1.77. This has been assigned by the DHCP server through the access point. That service automatically assigned a private IP address to my machine so it can be recognized in network communication. See: Figure 3.

```
crawfodz@ad.jmu.edu@L2020AIO12:~$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST>  mtu 1500
        inet 172.17.0.1  netmask 255.255.0.0  broadcast 172.17.255.255
        ether f2:fe:49:ea:96:2d  txqueuelen 0  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 44 overruns 0  carrier 0  collisions 0

enp0s31f6: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.1.77  netmask 255.255.255.0  broadcast 192.168.1.255
        inet6 fe80::6bdd:f944:349e:3d8  prefixlen 64  scopeid 0x20<link>
        ether 20:88:10:74:d2:09  txqueuelen 1000  (Ethernet)
        RX packets 159558  bytes 132182872 (132.1 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 40815  bytes 5189445 (5.1 MB)
        TX errors 0  dropped 15 overruns 0  carrier 0  collisions 0
        device interrupt 19  memory 0x74100000-74120000
```

Figure 3: IP address acquired from the AP

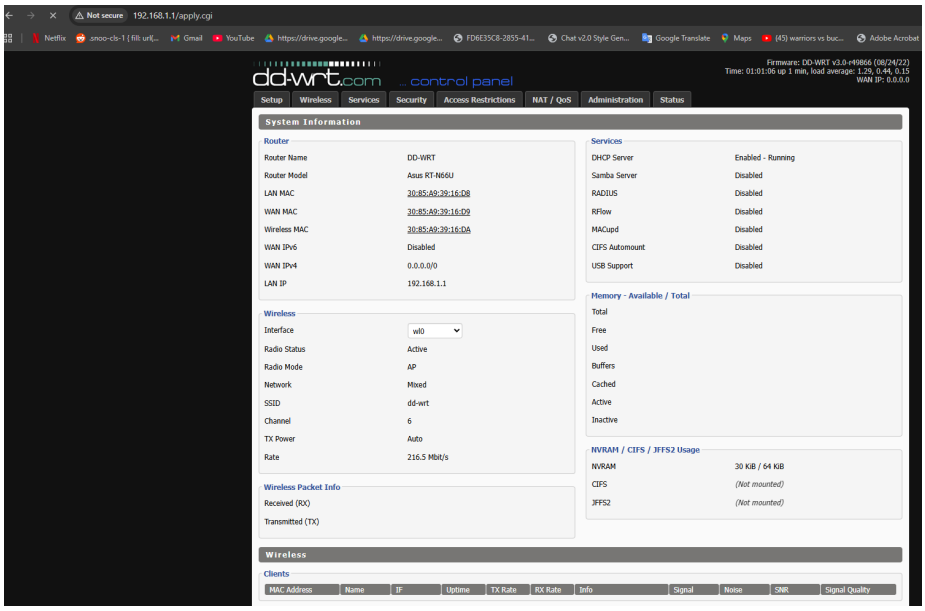


Figure 4: AP Login through my browser

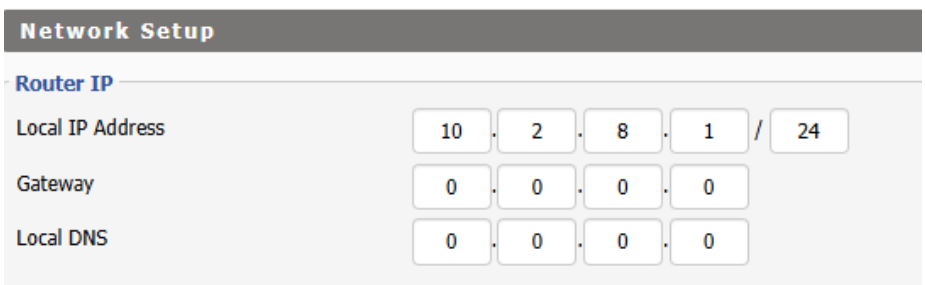


Figure 5: Changed Local IP Address

T3: Access Point (AP) Configuration

Q1: Is 10.2.5.1 (or your configured IP) a public or private address? Provide eveidence supporting your answer (output from ipcalc or other reliable sources).

A1: Our IP address 10.2.8.1 is a private IP address. We know this because ipcalc lists it as private in the terminal. See: Figure 6

```
crawfodz@ad.jmu.edu@L2020AIO12:~$ ipcalc 10.2.8.1
Address:   10.2.8.1             00001010.00000010.00001000. 00000001
Netmask:   255.255.255.0 = 24   11111111.11111111.11111111. 00000000
Wildcard:  0.0.0.255            00000000.00000000.00000000. 11111111
=>
Network:   10.2.8.0/24          00001010.00000010.00001000. 00000000
HostMin:   10.2.8.1             00001010.00000010.00001000. 00000001
HostMax:   10.2.8.254           00001010.00000010.00001000. 11111110
Broadcast: 10.2.8.255           00001010.00000010.00001000. 11111111
Hosts/Net: 254                   Class A, Private Internet
```

Figure 6: New IP address details in ipcalc

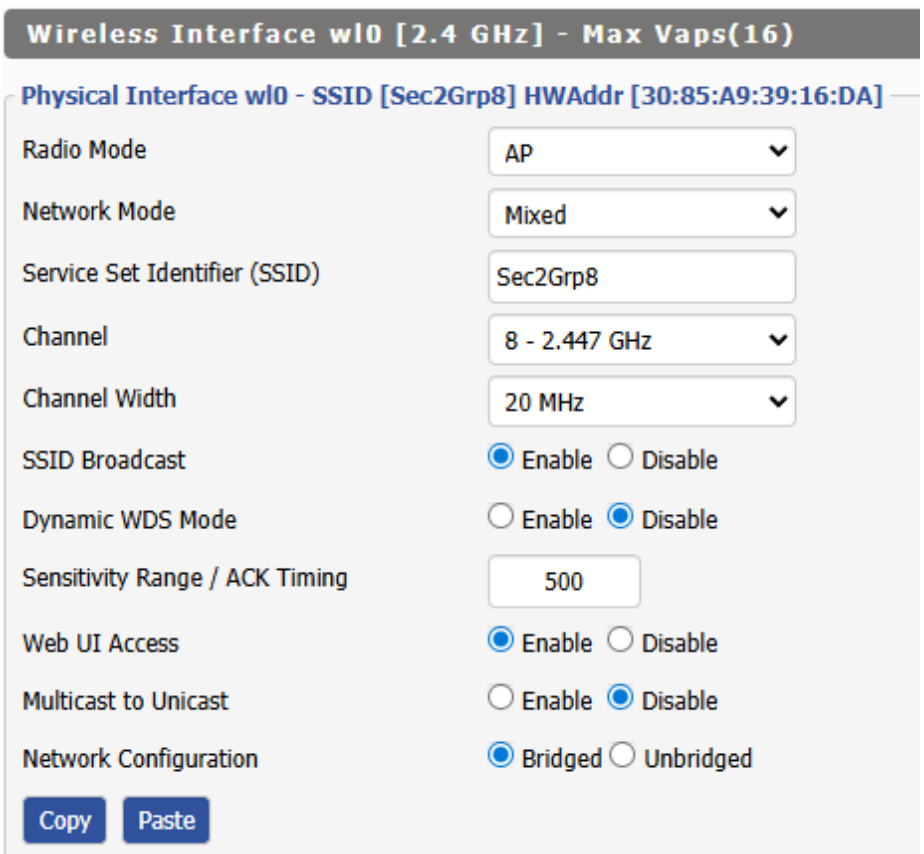


Figure 7: New Network Name and Wireless Channel

T4: Wireless Setting Update

Q1: What is the new IP address acquired by your desktop after reconfiguring the AP? Document with a screenshot.

A1: The newly acquired IP address is 10.2.5.77. See: Figure 8.

```
enp0s31f6: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.2.8.77  netmask 255.255.255.0  broadcast 10.2.8.255
        inet6 fe80::6bdd:f944:349e:3d8  prefixlen 64  scopeid 0x20<link>
        ether 20:88:10:74:d2:09  txqueuelen 1000  (Ethernet)
        RX packets 195587  bytes 157375194 (157.3 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 71275  bytes 8080957 (8.0 MB)
        TX errors 0  dropped 90 overruns 0  carrier 0  collisions 0
        device interrupt 19  memory 0x74100000-74120000
```

Figure 8: New Specific IP Address

T5: Inter-Host Connectivity Testing

Q1: Was your ping successful? If not, explain why inter-host connectivity failed at this stage.

A1: Inter-host connectivity failed becauuse at this point the AP has not established the ability to connect to other devices on a network, it has only established it's own public IP which could be used for communication. See figure: 9.

```
crawfodz@ad.jmu.edu@L2020AI012:~$ ping -c 5 192.168.99.231
PING 192.168.99.231 (192.168.99.231) 56(84) bytes of data.
From 10.2.8.1 icmp_seq=1 Destination Net Unreachable
From 10.2.8.1 icmp_seq=2 Destination Net Unreachable
From 10.2.8.1 icmp_seq=3 Destination Net Unreachable
From 10.2.8.1 icmp_seq=4 Destination Net Unreachable

--- 192.168.99.231 ping statistics ---
5 packets transmitted, 0 received, +4 errors, 100% packet loss, time 4057ms
```

Figure 9: Failed External Ping

Q1: What is the WAN IP address assigned to the AP's WAN port? Is it in a private or public range? How did you find it?

A1: The IP address assigned to the AP's WAN port was 19.168.99.180/23. This is shown as a private address in ipcalc. The WAN IP was found in the WAN tab under status while logged into the AP. See figure: 10.

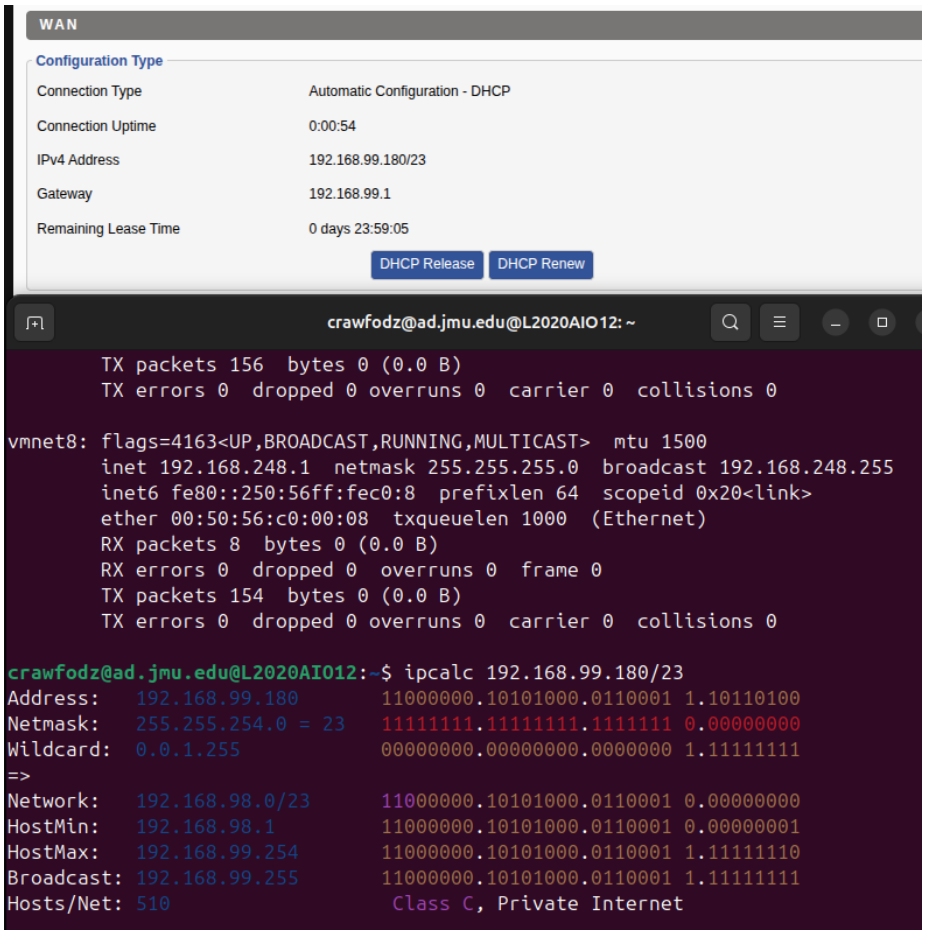


Figure 10: WAN IP Information

T6: End-to-End Connectivity

Q1: Is the ping to the external server successful? Explain the reason for the observed result based on the NAT and routing.

A1: The ping to the external server (google) was now successful. This is due to the AP now having a connection to a broader network. Earlier, the AP was only connected to the linux machine, making other connections impossible. But now that the WAN port is now connected to the lab switch, the AP and linux machine now have access to public domains. See figure: 11.

```
crawfodz@ad.jmu.edu@L2020AIO12:~$ ping  -c 5 google.com
PING google.com (142.251.111.102) 56(84) bytes of data.
64 bytes from bk-in-f102.1e100.net (142.251.111.102): icmp_seq=1 ttl=106 time=6.
91 ms
64 bytes from bk-in-f102.1e100.net (142.251.111.102): icmp_seq=2 ttl=106 time=6.
95 ms
64 bytes from bk-in-f102.1e100.net (142.251.111.102): icmp_seq=3 ttl=106 time=6.
88 ms
64 bytes from bk-in-f102.1e100.net (142.251.111.102): icmp_seq=4 ttl=106 time=6.
36 ms
64 bytes from bk-in-f102.1e100.net (142.251.111.102): icmp_seq=5 ttl=106 time=6.
70 ms

--- google.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 6.360/6.760/6.952/0.217 ms
crawfodz@ad.jmu.edu@L2020AIO12:~$
```

Figure 11: Successful ping to google.com

T7: Network Diagram

Q1: Draw a labeled network diagram showing all the hosts, IP assignments (LAN and WAN), AP, and any relevant connections. Submit this diagram in your lab report.

A1: See figure: 12.

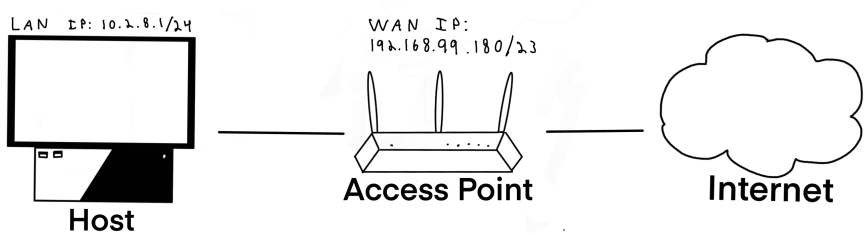


Figure 12: Connection Diagram

2.2 Exercise 2 - Wireless LAN Setup, Iperf Measurement, and Wireshark Analysis

T1: Network Configuration

Q1: Provide a network diagram and screenshots of each machine's IP configuration and wireless connection status.

A1: DHCP is automatically assigning IP addresses to the machines, but it is in the range of the switch that we specified. They are listed as active clients in the switch control center. See: Figure 13 and Figure 14

.

DHCP Clients

Hostname	IP Address	MAC Address	Lease Expiration	Static Leases
L2020AIO24	10.0.8.241	58:6D:67:19:9D:85	0 days 23:55:29	
Victorias-MBP	10.0.8.105	6C:7E:67:CA:95:A7	0 days 23:57:23	
L2020AIO25	10.0.8.135	A8:3C:A5:21:64:AD	0 days 23:49:12	

Figure 13: IP Addressed for Connected Machines

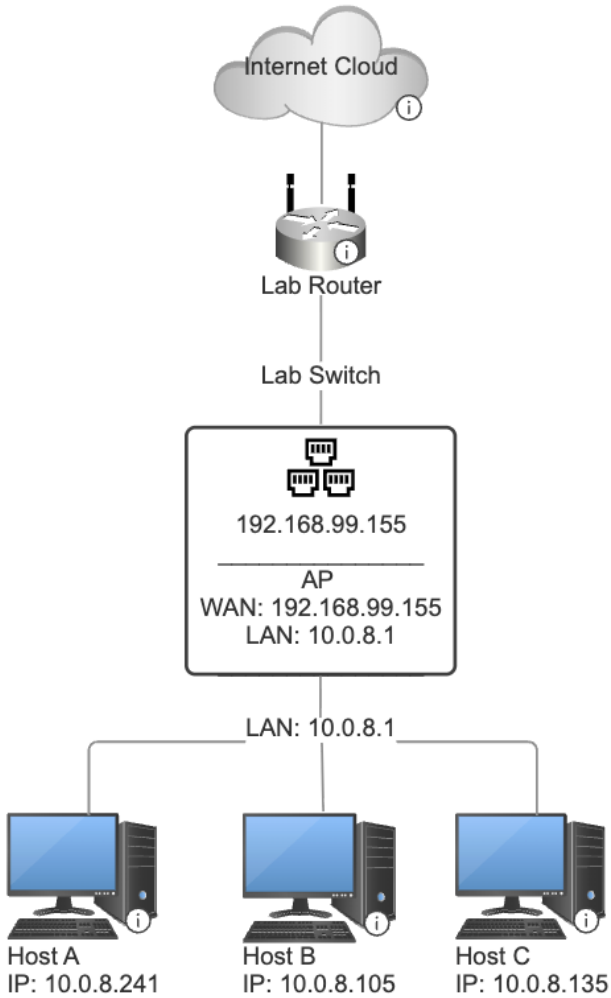


Figure 14: IP Configuration Diagram

T2: Baseline Network Measurement Using Iperf

Q1: Attach table(s) or screenshots summarizing your Iperf TCP and UDP results. What do you observe about bandwidth, jitter, and loss in each case?

A1: It is interesting to see that the bandwidth and transfer speeds of the TCP transfer packet was actually faster than the UDP packet. And the UDP connection did not recieve the last datagram after 10 attempts, which is a side affect of a connectionless protocol. TCP was the better protocol all around in this situation because it was faster and more reliable. See: Figure 15

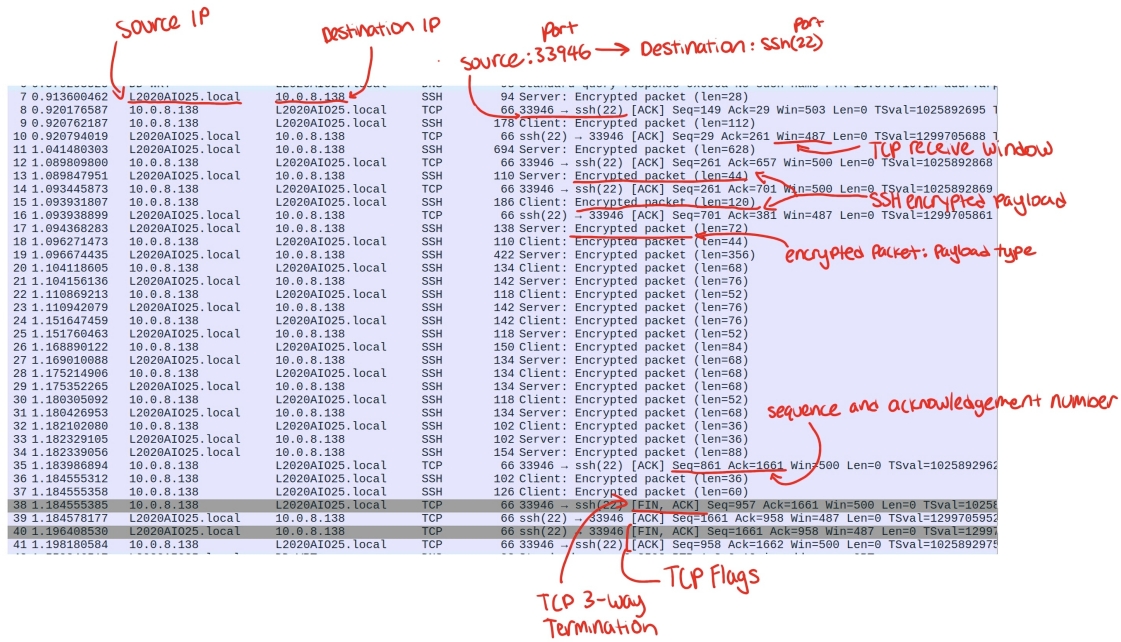
```
crawfodz@ad.jmu.edu@L2020AIO24:~$ iperf -c 10.0.8.13
------------------------------------------------------------
Client connecting to 10.0.8.13, TCP port 5001
TCP window size: 16.0 KByte (default)
------------------------------------------------------------
[  1] local 10.0.8.138 port 36814 connected with 10.0.8.13 port 5001
irtt=14/1448/114794)
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-11.6788 sec  31.4 MBytes  22.5 Mbits/sec
crawfodz@ad.jmu.edu@L2020AIO24:~$ iperf -c 10.0.8.13 -u
------------------------------------------------------------
Client connecting to 10.0.8.13, UDP port 5001
Sending 1470 byte datagrams, IPG target: 11215.21 us (kalman adjust)
UDP buffer size:  208 KByte (default)
------------------------------------------------------------
[  1] local 10.0.8.138 port 35440 connected with 10.0.8.13 port 5001
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-10.0153 sec  1.25 MBytes  1.05 Mbits/sec
[  1] Sent 896 datagrams
read failed: Connection refused
read failed: Connection refused
[  3] WARNING: did not receive ack of last datagram after 10 tries.
crawfodz@ad.jmu.edu@L2020AIO24:~$
```

Figure 15: TCP and UDP iperf results

T3: Generating and Capturing Network Traffic

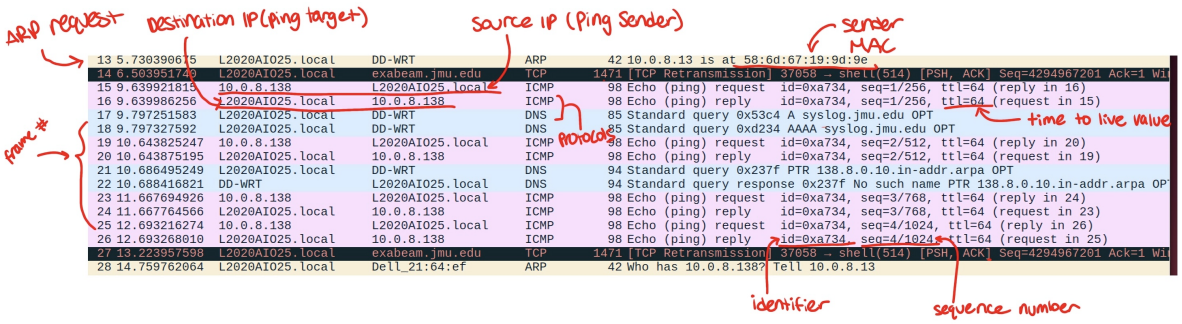
Q1: Provide annotated Wireshark screenshots showing successful capture of ICMP (ping), TCP/UDP (Iperf), and file-sharing packets on the wireless network. Highlight relevant protocol fields (e.g., MAC addresses, IP addresses, frame types).

A1: ***See: Figure 16 and Figure 17



7	0.913600462	L2020AIO25.local	10.0.8.138	SSH	94	Server: Encrypted packet (len=28)
8	0.920176587	10.0.8.138	L2020AIO25.local	TCP	66	33946 → ssh(22) [ACK] Seq=149 Ack=29 Win=503 Len=0 TSval=1025892695 T
9	0.920762187	10.0.8.138	L2020AIO25.local	SSH	178	Client: Encrypted packet (len=112)
10	0.920794019	L2020AIO25.local	10.0.8.138	TCP	66	ssh(22) → 33946 [ACK] Seq=29 Ack=261 Win=487 Len=0 TSval=1299705688 T
11	1.041480303	L2020AIO25.local	10.0.8.138	SSH	694	Server: Encrypted packet (len=628)
12	1.089809800	10.0.8.138	L2020AIO25.local	TCP	66	33946 → ssh(22) [ACK] Seq=261 Ack=657 Win=500 Len=0 TSval=1025892868
13	1.089847951	L2020AIO25.local	10.0.8.138	SSH	110	Server: Encrypted packet (len=44)
14	1.093445873	10.0.8.138	L2020AIO25.local	TCP	66	33946 → ssh(22) [ACK] Seq=261 Ack=701 Win=500 Len=0 TSval=1025892869
15	1.093931807	10.0.8.138	L2020AIO25.local	SSH	186	Client: Encrypted packet (len=120)
16	1.093938899	L2020AIO25.local	10.0.8.138	TCP	66	ssh(22) → 33946 [ACK] Seq=701 Ack=381 Win=487 Len=0 TSval=1299705861
17	1.094368283	L2020AIO25.local	10.0.8.138	SSH	138	Server: Encrypted packet (len=72)
18	1.096271473	10.0.8.138	L2020AIO25.local	SSH	110	Client: Encrypted packet (len=44)
19	1.096674435	L2020AIO25.local	10.0.8.138	SSH	422	Server: Encrypted packet (len=356)
20	1.104118605	10.0.8.138	L2020AIO25.local	SSH	134	Client: Encrypted packet (len=68)
21	1.104156136	L2020AIO25.local	10.0.8.138	SSH	142	Server: Encrypted packet (len=76)
22	1.110869213	10.0.8.138	L2020AIO25.local	SSH	118	Client: Encrypted packet (len=52)
23	1.110942079	L2020AIO25.local	10.0.8.138	SSH	142	Server: Encrypted packet (len=76)
24	1.151647459	10.0.8.138	L2020AIO25.local	SSH	142	Client: Encrypted packet (len=76)
25	1.151760463	L2020AIO25.local	10.0.8.138	SSH	118	Server: Encrypted packet (len=52)
26	1.168890122	10.0.8.138	L2020AIO25.local	SSH	150	Client: Encrypted packet (len=84)
27	1.169010088	L2020AIO25.local	10.0.8.138	SSH	134	Server: Encrypted packet (len=68)
28	1.175214906	10.0.8.138	L2020AIO25.local	SSH	134	Client: Encrypted packet (len=68)
29	1.175352265	L2020AIO25.local	10.0.8.138	SSH	134	Server: Encrypted packet (len=68)
30	1.180305092	10.0.8.138	L2020AIO25.local	SSH	118	Client: Encrypted packet (len=52)
31	1.180426953	L2020AIO25.local	10.0.8.138	SSH	134	Server: Encrypted packet (len=68)
32	1.182102080	10.0.8.138	L2020AIO25.local	SSH	102	Client: Encrypted packet (len=36)
33	1.182329105	L2020AIO25.local	10.0.8.138	SSH	102	Server: Encrypted packet (len=36)
34	1.182339056	L2020AIO25.local	10.0.8.138	SSH	154	Server: Encrypted packet (len=88)
35	1.183986894	10.0.8.138	L2020AIO25.local	TCP	66	33946 → ssh(22) [ACK] Seq=861 Ack=1661 Win=500 Len=0 TSval=1025892962
36	1.184555312	10.0.8.138	L2020AIO25.local	SSH	102	Client: Encrypted packet (len=36)
37	1.184555358	10.0.8.138	L2020AIO25.local	SSH	126	Client: Encrypted packet (len=60)
38	1.184555385	10.0.8.138	L2020AIO25.local	TCP	66	33946 → ssh(22) [FIN, ACK] Seq=957 Ack=1661 Win=500 Len=0 TSval=10258
39	1.184578177	L2020AIO25.local	10.0.8.138	TCP	66	ssh(22) → 33946 [ACK] Seq=1661 Ack=958 Win=487 Len=0 TSval=1299705952
40	1.196408530	L2020AIO25.local	10.0.8.138	TCP	66	ssh(22) → 33946 [FIN, ACK] Seq=1661 Ack=958 Win=487 Len=0 TSval=12997
41	1.198180584	10.0.8.138	L2020AIO25.local	TCP	66	33946 → ssh(22) [ACK] Seq=958 Ack=1662 Win=500 Len=0 TSval=1025892975

Figure 16: Wireshark Annotations Pt.1



13	5.730390615	L2020AIO25.local	DD-WRT	ARP	42	10.0.8.13 is at 58:6d:67:19:9d:9e
14	6.593951740	L2020AIO25.local	exabeam.jmu.edu	TCP	1471	[TCP Retransmission] 37058 → shell(514) [PSH, ACK] Seq=4294967201 Ack=1 Wi
15	9.639921815	10.0.8.138	L2020AIO25.local	ICMP	98	Echo (ping) request id=0xa734, seq=1/256, ttl=64 (reply in 16)
16	9.639986256	L2020AIO25.local	10.0.8.138	ICMP	98	Echo (ping) reply id=0xa734, seq=1/256, ttl=64 (request in 15)
17	9.797251583	L2020AIO25.local	DD-WRT	DNS	85	Standard query 0x53c4 A syslog.jmu.edu OPT
18	9.797327592	L2020AIO25.local	DD-WRT	DNS	85	Standard query 0xd234 AAAA syslog.jmu.edu OPT
19	10.643825247	10.0.8.138	L2020AIO25.local	ICMP	98	Echo (ping) request id=0xa734, seq=2/512, ttl=64 (reply in 20)
20	10.643875195	L2020AIO25.local	10.0.8.138	ICMP	98	Echo (ping) reply id=0xa734, seq=2/512, ttl=64 (request in 19)
21	10.686495249	L2020AIO25.local	DD-WRT	DNS	94	Standard query 0x237f PTR 138.8.0.10.in-addr.arpa OPT
22	10.688416821	DD-WRT	L2020AIO25.local	DNS	94	Standard query response 0x237f No such name PTR 138.8.0.10.in-addr.arpa OPT
23	11.667694926	10.0.8.138	L2020AIO25.local	ICMP	98	Echo (ping) request id=0xa734, seq=3/768, ttl=64 (reply in 24)
24	11.667764566	L2020AIO25.local	10.0.8.138	ICMP	98	Echo (ping) reply id=0xa734, seq=3/768, ttl=64 (request in 23)
25	12.693216274	10.0.8.138	L2020AIO25.local	ICMP	98	Echo (ping) request id=0xa734, seq=4/1024, ttl=64 (reply in 26)
26	12.693268010	L2020AIO25.local	10.0.8.138	ICMP	98	Echo (ping) reply id=0xa734, seq=4/1024, ttl=64 (request in 25)
27	13.223957598	L2020AIO25.local	exabeam.jmu.edu	TCP	1471	[TCP Retransmission] 37058 → shell(514) [PSH, ACK] Seq=4294967201 Ack=1 Wi
28	14.759762064	L2020AIO25.local	Dell_21:64:ef	ARP	42	Who has 10.0.8.138? Tell 10.0.8.13

Figure 17: Wireshark Annotations Pt.2

3 Conclusion

3.1 Summary

This lab provided a deeper dive into configuring IP addresses and connections. Starting with a private network that only my client computer had access to, understanding its protocols, then using what was learned to establish connectivity and test connectivity with other machines was valuable. It also showed us different tools for finding IP information and different ways to test it. Examples were pinging a machine, sending a file using the terminal, and logging into the AP. Overall this was an eye opening and valuable lab that showed the different things we can use to evaluate an IP and the surrounding network.

3.2 Expectancy-Value-Cost

Expectancy
I felt I was well equipped to get the lab done. I did need some slight teacher assistance in the beginning with troubleshooting, but once that was taken care of the lab went mostly smooth.
Value
I think this lab was extremely valuable because we had some some tasks we had never done before and got much more into the specifics of IP addresses and how they are established along with how to use IP addresses to establish connections over the network to complete different tasks.
Cost
We put in around 7 hours into this lab.

3.3 Suggestions

Instruction - Do not include the question in your lab report.
Overall it was a straightforward lab.

3.4 Best Practices

Some best practices would be to read the questions carefully to fully understand what the problem is asking, and document step by step procedures.

4 Lab References

[1] OpenAI, *YouTube Tutorial: How to use SCP*. Accessed: Dec. 2, 2025. Available: https://youtu.be/gGe1cNgfHwE?si=lH9vpp2Dtdxoucsi

5 Acknowledgments

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